INNOVATOR TEST 4

MATH MODULE 1

1. $$-6x + 3 < y$$
$$y < x + 6$$

If (x, y) is a solution to the given system of inequalities, which of the following could be (x, y)?

PREV. QUESTION | NEXT QUESTION

A $(-2, -4)$

B $(-2, 1)$

C $(-2, 4)$

D $(4, 7)$

2. If $x = 9$, what is the value of $x + 23$?

PREV. QUESTION | NEXT QUESTION

A 14

B 23

C 32

D 34

3. Tires are tested by sampling 24 tires from each batch of 600 tires. If 87.5% of the tires in a sample pass inspection, how many tires in the entire batch would most likely pass inspection?

PREV. QUESTION | NEXT QUESTION

A 525

B 550

C 575

D 579

4. A square has a side length of 16 centimeters. What is the area, in square centimeters, of the square?

PREV. QUESTION | NEXT QUESTION

- A 4
- B 16
- C 64
- D 256

5. Right triangles ABC and DEF are similar, where A corresponds to D. Angle E is a right angle and the measure of angle C is 21°. What is the measure of angle D?

PREV. QUESTION | NEXT QUESTION

- A 21°
- B 69°
- C 79°
- D 159°

6. In a project for business class, Bobby launched a website with pop culture content and aimed to get as many visitors as possible. On day 6 there were 20 visitors to the website, and on day 18 there were 320 visitors. Which of the following best describes the average change in visitors to the website over the given period?

PREV. QUESTION | NEXT QUESTION

- A On average, the number of visitors to the website increased by 16 visitors per day
- B On average, the number of visitors to the website increased by 25 visitors per day.
- C On average, the number of visitors to the website increased by 18.25 visitors per day.
- D On average, the number of visitors to the website increased by 53.33 visitors per day.

7. Which expression is equivalent to $(5xy + 3x)(x - 2y)$?

PREV. QUESTION | NEXT QUESTION

- A $5x^2y - 6xy$
- B $5x^2y + 3x^2 + 10xy^2 - 6xy$
- C $5x^2y + 3x^2 - 10xy^2 - 6xy$
- D $-5x^2y^2 + 3x^2 - 6xy$

8. For the function $g(x) = x^2 - 5x - 14$, what value of x satisfies $g(x) = 0$, given that $x > 0$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

9. Data set X: $4, 5, 10, 10, 13, 15, 20$

Data set X is listed above. Which of the following is the difference between the mean and the median of the data set?

PREV. QUESTION | NEXT QUESTION

- A 0.5
- B 1
- C 7
- D 21

10. $-5x + 40kx = 120$

In the given equation, k is a constant. The equation has no solution. What is the value of k?

PREV. QUESTION | NEXT QUESTION

× A 0

× B $\frac{1}{8}$

× C $\frac{3}{5}$

× D 3

11. If $7(x - 11) = -17$, what is the value of x?

PREV. QUESTION | NEXT QUESTION

× A $-\frac{94}{7}$

× B $-\frac{24}{7}$

× C $-\frac{6}{7}$

× D $\frac{60}{7}$

12.

x	$f(x)$
6	−2
8	3
11	10.5

The table shows three values of x and their corresponding values for $f(x)$. The linear relationship between x and $f(x)$ is defined by the equation $f(x) = ax - 17$, where a is a constant. What is the value of a?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

13. $$c = \frac{23}{d} + 11$$

In the given equation, $d \neq 0$ and $c > 0$. Which of the following correctly expresses d in terms of c?

PREV. QUESTION | NEXT QUESTION

× A $d = 23c - 11$

× B $d = \frac{23}{c} - 11$

× C $d = \frac{23}{c-11}$

× D $d = \frac{34}{c}$

14. An angle has a measure of $\frac{13}{9}\pi$ radians. What is the measure of the angle in degrees?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

15. $$6x - 4y = 14$$
$$2x - 3y = 3$$

In the given system of equations, (x, y) is the solution. What is $x + y$?

PREV. QUESTION | NEXT QUESTION

× A -3

× B 1

× C 4

× D 8

16.

	Fewer than 3 hours	3 to 6 hours	More than 6 hours	Total
Class 1	6	19	5	30
Class 2	8	15	7	30
Total	14	34	12	60

A teacher asked the students in two of his classes how many hours they spent studying for their recent midterm. The results are shown in the table above. A student who studied 6 hours or fewer is randomly chosen. What is the probability the student is from Class 1?

PREV. QUESTION NEXT QUESTION

× A $\frac{5}{30}$

× B $\frac{5}{12}$

× C $\frac{25}{48}$

× D $\frac{25}{30}$

17. The expression $\frac{40}{8x+56}$ is equivalent to $\frac{5}{x+k}$, where k is a constant and $x \neq -k$. What is the value of k?

PREV. QUESTION NEXT QUESTION

× A 7

× B 8

× C 35

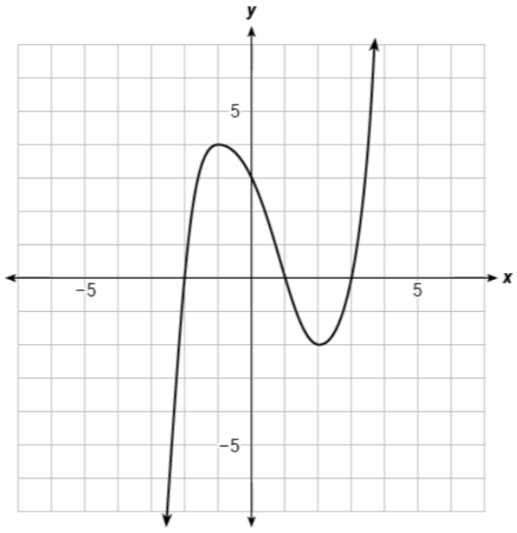
× D 56

18. Andy and Jen plan to save a combined $\$420$ each month, and Jen will contribute $\$80$ more than Andy each month. How much money (in dollars) will Andy have contributed after six months?

PREV. QUESTION NEXT QUESTION

COME BACK LATER

19.



A polynomial function $p(x)$ is graphed on the xy-plane. What is the equation for the function $p(x)$?

PREV. QUESTION | NEXT QUESTION

× A $p(x) = -0.5(x+2)(x-1)(x-3)$

× B $p(x) = -0.5(x-2)(x+1)(x+3)$

× C $p(x) = 0.5(x+2)(x-1)(x-3)$

× D $p(x) = 0.5(x-2)(x+1)(x+3)$

COME BACK LATER

20. A line segment in the xy-coordinate plane has endpoints $A(1,3)$ and $B(5,1)$. Which of the following is the equation of a line perpendicular to segment $\overline{AB}$ and intersects $\overline{AB}$ at its midpoint?

PREV. QUESTION | NEXT QUESTION

× A $y = \frac{1}{2}x + \frac{1}{2}$

× B $y = \frac{1}{2}x + \frac{5}{2}$

× C $y = 2x - 1$

× D $y = 2x - 4$

21. What is the perimeter of an isosceles right triangle where each leg has a length of $12\sqrt{5}$?

PREV. QUESTION | NEXT QUESTION

× A $12\sqrt{5} + 12\sqrt{10}$

× B $24\sqrt{5} + 12\sqrt{10}$

× C $24\sqrt{15}$

× D $36\sqrt{15}$

22. In the equation $x^2 - 21x + k = 0$, k is a constant. What is the least possible integer value of k for which this equation has no real solutions?

PREV. QUESTION

COME BACK LATER

MODULE 2 (HARDER)

1. If $|7x + 14| = 98$ and x is positive, what is the value of $x + 2$?

NEXT QUESTION

COME BACK LATER

2.

a	3	6	9	12
$f(a)$	−2	−1	0	1

The table above shows some values of the linear function f. Which of the following defines f?

PREV. QUESTION NEXT QUESTION

A $f(a) = \frac{1}{3}a - 3$

B $f(a) = -\frac{1}{3}a - 1$

C $f(a) = a - 5$

D $f(a) = 2a - 8$

3. Stan's plant grows 2.5 inches per week. It is 15 inches tall now. In how many weeks will the plant be 42.5 inches tall?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

4. $$3(x-2)-2(x-1)=-x+2(x+2)-8$$

How many solutions does the given equation have?

PREV. QUESTION | NEXT QUESTION

A Zero

B Exactly one

C Exactly two

D Infinitely many

5. $$f(x+3)=\sqrt{x^3}+17$$

For the given function, what is the value of $f(4)$?

PREV. QUESTION | NEXT QUESTION

A 18

B 25

C 28

D 35.5

6. Theresa invested in a company, and the value of her investment increased by 9% from 2021 to 2022. If the value of her investment in 2022 is k times the value it was in 2021, what is the value of k?

PREV. QUESTION | NEXT QUESTION

× A 0.09

× B 0.9

× C 1.09

× D 1.9

7. Which of the following expressions can equal -1 for some value of a?

PREV. QUESTION | NEXT QUESTION

× A $|a-3|+2$

× B $|a-2|+1$

× C $|a-1|$

× D $|a-2|-2$

8. $$\frac{m}{n} = 3$$
$$6(n-4) = m$$

Given the system of equations above, what is the value of n?

PREV. QUESTION | NEXT QUESTION

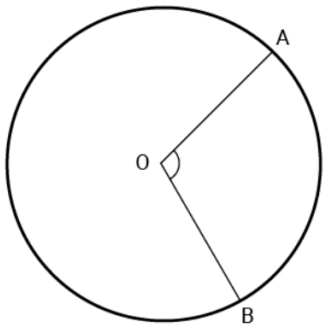
× A 4

× B 8

× C 16

× D 24

9.



In circle O, the length of AO is 12 units and the length of minor arc AB is 8π. What is the measure, in degrees, of $\angle AOB$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

10. Which expression is equivalent to $\frac{x+4}{2x+2} + \frac{5}{x+1}$?

PREV. QUESTION | NEXT QUESTION

A $\frac{x+7}{x+1}$

B $\frac{x+9}{3x+3}$

C $\frac{x+14}{2x+2}$

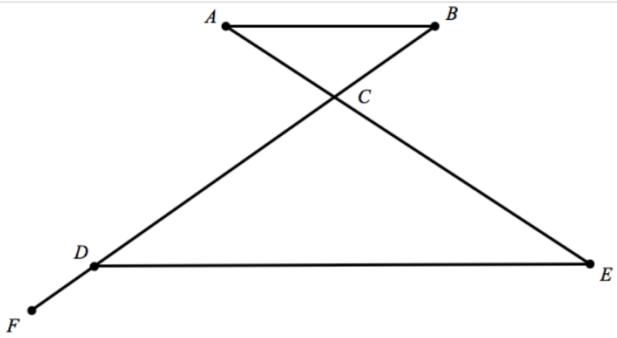
D $\frac{5x+20}{2x^2+4x+2}$

11. Which of the following is the equation of $g(x) = 2x$ when it is moved 1 unit to the left and 1 unit up?

PREV. QUESTION | NEXT QUESTION

- A $y = 2x - 1$
- B $y = 2x + 1$
- C $y = 2x + 5$
- D $y = 2x + 3$

12.



If $\overline{AB} \parallel \overline{DE}$, $AC = BC$, and the measure of $\angle ACB = 88^\circ$, what is $m\angle FDE$?

PREV. QUESTION | NEXT QUESTION

- A 88°
- B 92°
- C 128°
- D 134°

13. The function $f(x) = mx + b$ and the function $g(x) = nx + b$ are defined for all real values of x. Which statement is true about the functions?

PREV. QUESTION | NEXT QUESTION

× A If $|m| > |n|$, then $f(x) > g(x)$ for all values of x.

× B If $|m| = |n|$, then $f(x) = g(x)$ for all values of x.

× C If $m > 0$, then $f(x) > g(x)$ for all values of x.

× D If $m \neq n$, then $f(x) = g(x)$ for only one value of x.

14. Triangles ABC and BCD are right triangles, where $\angle A$ and $\angle D$ are right angles. If $AB = AC$ and $BD = CD$, what is the value of $tan(\angle ABC) + tan(\angle DBC)$?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

15. The first term in a sequence is 21. Each term after the first term is 1.5 times the preceding term. If s represents the nth term of the sequence, which equation gives s in terms of n?

PREV. QUESTION | NEXT QUESTION

× A $s = 1.5(21^n)$

× B $s = 1.5(21^{n-1})$

× C $s = 21(1.5^n)$

× D $s = 21(1.5^{n-1})$

16.

Vocabulary Test Scores	
Test Score	Number of Students
100	1
90	5
80	10
70	8
60	2

The table above shows the distribution of test scores in Mrs. Keaton's English class. Which of the following sets of test scores, if added, would change the standard deviation of the distribution the least?

PREV. QUESTION | NEXT QUESTION

× A $60, 60, 60$

× B $70, 80, 90$

× C $60, 100, 100$

× D $90, 90, 90$

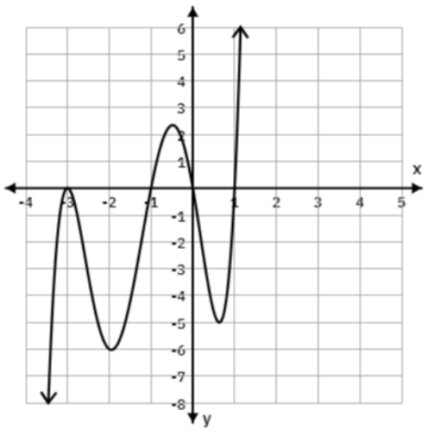
COME BACK LATER

17. The graph of $2y - 2x^2 + 14x = -36$ is translated to the left 3 units in the xy-plane. What is the positive x-coordinate of the x-intercept of the resulting graph?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

18.



The graph of $y = f(x)$ is shown. Which of the following best represents the function of $f(x)$?

PREV. QUESTION | NEXT QUESTION

- A $f(x) = x(x-1)(x+1)(x+3)^2$
- B $f(x) = x(x+1)(x-1)(x+3)$
- C $f(x) = (x-3)^2(x-1)(x+1)$
- D $f(x) = (x+3)^2(x+1)(x-1)$

COME BACK LATER

19. Which of the following expressions is equivalent to $\frac{x+3}{x^2-4x+3} + \frac{2}{x-1}$?

PREV. QUESTION | NEXT QUESTION

- A $\frac{1}{x-1}$
- B $\frac{3}{x-3}$
- C $\frac{3x+3}{x^2-4x+3}$
- D $\frac{x+5}{x^2-3x+2}$

20.

$$3x + by = 36$$

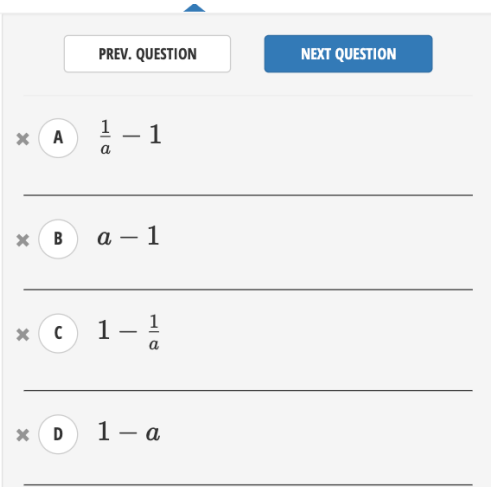
$$ax - 7y = 24$$

If the given system of equations has no solution, what is the value of ab?

PREV. QUESTION | NEXT QUESTION

COME BACK LATER

21. The point (a, a) satisfies the linear inequality $y \leq mx + 1$, and $a > 0$. The value of m must be greater than or equal to what value?



22. A ball is kicked off of a platform modeled by the equation $y = -5x^2 + bx + 10$, such that y is height in feet, x is seconds, and b is a constant. The ball reaches a maximum height of 11.25 feet, and it takes 2 seconds to hit the ground. How many seconds does it take for the ball to reach its maximum height?

